PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Long term ocular and visual outcomes following symptomatic and asymptomatic congenital CMV infection: a systematic review protocol
AUTHORS	Karamchandani, Urvi; Ahmed, Umar; Rufai, Sohaib; Tan, Naomi; Tan, Weijen; Petrushkin, Harry; Solebo, Ameenat

VERSION 1 – REVIEW

REVIEWER	Demmler, Gail Texas Children's Hospital, Pediatric Infectious Diseases
REVIEW RETURNED	13-Dec-2021

GENERAL COMMENTS	The authors propose to perform a protocol of systematic review of ocular and visual outcomes in congenital CMV infection; there is a paucity of reliable information on this important outcome so the authors may be able to provide information on this knowledge gap, and provide the information in a summary that will be useful to clinicians and to research scientists alike. However, they may find there are not very many studies on this outcome that are well done, so they may need to adjust for that. The authors may consider shortening the introduction a little and focus on visual outcomes, and less on auditory sensorineural hearing outcomes, to streamline the manuscript development.

REVIEWER	Keenan, Jeremy University of California, San Francisco, F.I. Proctor Foundation
REVIEW RETURNED	08-Feb-2022

GENERAL COMMENTS	This is a sound protocol for a systematic review on vision and ophthalmic outcomes among children with congenital CMV. Just a few minor points that might make it clearer: P9, line 12: I think when searching “cytomegalovirus” that Pubmed is smart enough to also include “CMV”. But I am not entirely sure. Just flagging for consideration if you should add CMV to the search terms or if doing so would be redundant for all those databases. (Might be a question for the research librarian.) P9, line 45: I usually think of longitudinal studies being cohort studies. Is number 4 on this list meant to communicate that case control studies would be excluded? What about a secondary analysis of a randomized trial? (Although I suppose that technically could be considered a longitudinal study.) Do you need to exclude any study designs at all? It seems to me there will be very few studies in general on this topic so might not need a lot of exclusion criteria?

	P11, line 40: please be more clear about outcomes. Is it 4 separate outcomes? (eg, vision impairment incidence, vision impairment prevalence, ophthalmic disorder incidence, ophthalmic disorder prevalence) or one joint outcome? (prevalence or incidence of vision impairment or ophthalmic disorder)— I see later down that you will report separate analyses for prevalence and incidence, but still might be nice to be clearer in this section? L11, line 40: are you pre-specifying definitions for what constitutes "vision impairment" and what constitutes an "ophthalmic disorder"? Or are you purposefully leaving this a bit vague in the anticipation that different studies will have different reporting conventions, etc (which I think is reasonable; you might just want to call this out in that case?) Because there are many things that could fit into these definitions. I imagine meta-analyses may be difficult due to study heterogeneity, but if you do attempt meta-analyses, how will you decide on the final outcomes? Simply by seeing the most common outcomes used in the studies? Or by focusing on "vision-threatening" or "vision-impairing" outcomes? (I doubt there will be any reports of non-visually-significant conditions in these kids, but not sure you want to consider a non-sight-threatening disease to qualify as an "ophthalmic disorder"? It might be nice to at least indicate your plan for this? P12, line 6: "neurological versus other": what about asymptomatic or studies that don't mention symptoms? Will that be a separate group or lumped with "other" P14, line 3: maybe temper this statement with "typically not amenable…" These kids may get certain treatments. P14, line 12: does the "separate" here mean separate models for prevalence and incidence? Or separate models for symptomatic and asymptomatic kids? Or both. Based on the final paragraph of the paper it seems that it will be important to have separate analyses for the symptomatic and asymptomatic groups, but maybe this could be articulated better. (And also, it might be nice to say somewhere that descriptive analyses will also consider the differences between symptomatic and asymptomatic kids—which would be important in the case that meta-analyses are not possible due to low numbers or heterogeneity)

REVIEWER	Flores-Medina, Saúl Instituto Nacional de Perinatologia, Infectología
REVIEW RETURNED	09-Feb-2022

GENERAL COMMENTS	The aim and design of this study seems very interesting, it might provide the scientific community with important findings, since data about the effects of CMV infection on eye damage and impairment of vision are really scarce, mainly in developing countries.Furthermore, those findings will also be key for the development of guidelines and intervention strategies in ophthalmological care of CMV-infected patients, whether symptomatic of asymptomatic.The methods proposed in this study have been fully validated, so they are appropriate for this type of study, avoiding data biases and ensuring accuracy of the results.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1			
The authors may consider shortening the introduction a little and focus on visual outcomes, and less on auditory sensorineural hearing outcomes, to streamline the manuscript development.	Many thanks. Hearing loss as a result of congenital CMV is, within most countries, an indication for longer term surveillance to detect new eye problems, because of the significant additional burden of dual sensory loss ("This (detecting eye problems) is of particular importance for a population who are also at risk of cCMV related hearing impairment") We have not removed the section on hearing loss as it is key that the general audience understand the rationale for enhanced eye surveillance for this population even when infants appear asymptomatic. We have, however, shortened the section on hearing loss.	Removal of "necessitating regular audiological monitoring for these infants"	Line 129
Reviewer: 2			
P9, line 12: I think when searching "cytomegalovirus" that Pubmed is smart enough to also include "CMV". But I am not entirely sure. Just flagging for consideration if you should add CMV to the search terms or if doing so would be redundant for all those databases. (Might be a question for the research librarian.)	The full search strategy has ben provided and will include "CMV"	S1 search strategy	Appendix
P9, line 45: I usually think of longitudinal studies being cohort studies. Is number 4 on this list meant to communicate that case	Our outcome measure is frequency amongst an at-risk population. It would be challenging to get this information from a case control	No change	

control studies would be excluded? What about a secondary analysis of a randomized trial? (Although I suppose that technically could be considered a longitudinal study.) Do you need to exclude any study designs at all? It seems to me there will be very few studies in general on this topic so might not need a lot of exclusion criteria?	or RCT study unless there was longitudinal data collection.		
P11, line 40: please be more clear about outcomes. Is it 4 separate outcomes? (eg, vision impairment incidence, vision impairment prevalence, ophthalmic disorder incidence, ophthalmic disorder prevalence) or one joint outcome? (prevalence or incidence of vision impairment or ophthalmic disorder)— I see later down that you will report separate analyses for prevalence and incidence, but still might be nice to be clearer in this section?	rephrased, thank you	"primary outcome measures will be incidence and or prevalence of vision impairment or ophthalmic disorder in patients with…"	Line 11
L11, line 40: are you pre-specifying definitions for what constitutes "vision impairment" and what constitutes an "ophthalmic disorder"? Or are you purposefully leaving this a bit vague in the anticipation that different studies will have different reporting conventions, etc (which I think is reasonable; you might just want to call this out in that case?) Because there are many things that could fit into these definitions. I	We have defined the outcomes of interest in more detail	"The international classification of disease (ICD-11) definitions of visual impairment and blindness will be used to classify visual outcomes. Ophthalmic disorders will be defined as sight threatening structural or neuro-ophthalmic abnormalities" and "Outcome types (eg degree of severity of visual impairment, moderate, severe or	Line 209 - 214

imagine meta-analyses may be difficult due to study heterogeneity, but if you do attempt meta-analyses, how will you decide on the final outcomes? Simply by seeing the most common outcomes used in the studies? Or by focusing on "vision-threatening" or "vision-impairing" outcomes? (I doubt there will be any reports of non-visually-significant conditions in these kids, but not sure you want to consider a non-sight-threatening disease to qualify as an "ophthalmic disorder"? It might be nice to at least indicate your plan for this?		blindness; anatomical location of ocular abnormality)	
P12, line 6: "neurological versus other": what about asymptomatic or studies that don't mention symptoms? Will that be a separate group or lumped with "other"	We have specified that analyses will be done separately for the symptomatic and asymptomatic groups	"All analyses will be performed separately for symptomatic and asymptomatic cCMV." and "neurological versus other for symptomatic cCMV only"	Line 227, 222
P14, line 3: maybe temper this statement with "typically not amenable…" These kids may get certain treatments.	Requested change made	"Typically" added	Line 271
P14, line 12: does the "separate" here mean separate models for prevalence and incidence? Or separate models for symptomatic and asymptomatic kids? Or both. Based on the final paragraph of the paper it seems that it will be important to have separate analyses for the symptomatic and asymptomatic groups, but	As above, we have described that analyses will be performed separately	All analyses will be performed separately for symptomatic and asymptomatic cCMV.	Line 227

maybe this could be articulated better. (And also, it might be nice to say somewhere that descriptive analyses will also consider the differences between symptomatic and asymptomatic kids—which would be important in the case that meta-analyses are not possible due to low numbers or heterogeneity)			
Reviewer: 3			
We thank this reviewer for their comments and note that there were no recommended changes			

VERSION 2 – REVIEW

REVIEWER	Demmler, Gail Texas Children's Hospital, Pediatric Infectious Diseases
REVIEW RETURNED	28-Mar-2022

GENERAL COMMENTS	The authors describe a research plan for a systematic review of available literature evidence on ocular involvement and ocular sequelae associated with symptomatic and asymptomatic congenital CMV infection. The background of the issue and the pertinent knowledge gasps are presented, as is a rational data plan. There is a knowledge gap on long term ocular outcomes in congenital CMV, and the authors are likely to find a paucity of well done studies, and lead the way, hopefully, for more in depth studies on this important clinical issue. They may wish to include analysis of children with BOTH deafness/hearing impairment AND blindness, vision impairment as a separate multi sensory outcomes analysis, as they represent a specialized group of patients. Also, the authors may wish to see if there is a relationship between ocular involvement central nervous system involvement at birth and then long term. . Rarely is there isolated ocular findings in congenital CMV, so co morbid risks may be important to note, analyze, and comment upon.

REVIEWER	Keenan, Jeremy University of California, San Francisco, F.I. Proctor Foundation
REVIEW RETURNED	12-Apr-2022

GENERAL COMMENTS	The authors have addressed my comments.

VERSION 2 – AUTHOR RESPONSE

Reviewer: 1

Prof. Gail Demmler, Texas Children's Hospital, Baylor College of Medicine

Comments to the Author:

*(The authors) may wish to include analysis of children with BOTH deafness/hearing impairment AND blindness, vision impairment as a separate multi sensory outcomes analysis, as they represent a specialized group of patients.

Also, the authors may wish to see if there is a relationship between ocular involvement central nervous system involvement at birth and then long term.

Rarely is there isolated ocular findings in congenital CMV, so co morbid risks may be important to note, analyze, and comment upon.

Thank you – the primary evidence gap is around asymptomatic CMV, a gap which will be addressed by our review as a central aim.

We do however propose sub group analyses for the population with symptomatic CMV (“CMV symptom type (neurological versus other for symptomatic cCMV only)”, line 224) which should address the important aspects raised by the reviewer above.